

Review

A Survey on UxV Swarms and the Role of Artificial Intelligence as a Technological Enabler

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Abstract

In recent years, there has been an ever increasing interest in UxVs and the technology surrounding them. A more recent area of interest within the UxV ecosystem is the development of UxV swarms. In these systems, multiple UxVs synchronize, continuously exchange information, and operate as a cohesive unit. This evolution requires a higher level of autonomy, enhanced coordination, and more efficient communication channels. In this survey, we present relevant research on swarms of UxVs, always considering artificial intelligence (AI) as the key technological enabler for the swarm operations. We view the swarm from three distinct perspectives; these are intelligence-wise, communication-wise, and security-wise. Our main goal is to explore in which ways and to what extent AI has been integrated in these aspects. We aim to identify which of these aspects are the most researched and which need deeper investigation, the types of AI that are mainly used, and which types of vehicles are preferred. We then discuss the results of our work and present current limitations as well as areas of future research in the realm of UxVs, AI, swarm intelligence, communications, and security.

Keywords: swarm; UxV; UAV; UGV; UUV; drones; artificial intelligence; machine learning; swarm intelligence; communications; security



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1. Introduction

The rapid evolution of autonomous systems has fueled growing interest in integrating uncrewed vehicles (UxVs) across defense, civilian protection, research, logistics, agriculture, and environmental monitoring. These platforms span aerial (UAVs), ground (UGVs), and surface/underwater (USVs/UUVs) domains and offer unprecedented mobility and mission autonomy. Recent events in Ukraine have accelerated UAV development for military use [1], underscoring the inevitability of UxV adoption in additional sectors. This momentum is reflected in a large body of literature reviewing state-of-the-art (SotA) advances [2] while also proposing innovative applications, such as blockchain integration in underwater drones [3]. UAVs have great potential due to their broad applications [4–8], while unmanned marine vehicles (UMVs) are comparatively underexplored but show promise for industrial and civil protection scenarios [9–12]. Research on UGVs emphasizes distinct challenges in areas such as terrain adaptability and energy efficiency [13–15], with novel approaches in path planning [16]. Complementary studies explore cross-platform collaboration between UAVs and UGVs [17–19], highlighting the benefits of multimodal coordination. Beyond technical development, regulatory and governance issues are also examined [20–22]. Fur-

thermore, proposals for open-source drone technologies [23] illustrate ongoing debates about transparency and standardization.

As observed, the scientific community has often treated UxVs as individual assets, focusing primarily on their standalone capabilities and performance, as well as their applications in different domains, e.g., civil, military, commercial, etc. In this review, we move one step further and treat UxVs as parts of a swarm. This swarm comprises identical vehicles, all of which follow the same collaborative logic, communicate with the same protocols, and operate as nodes of a larger network. In addition, we want to explore how AI can enhance the autonomy and operational capabilities of the swarm and its individual units. Several surveys, such as [24,25], have examined AI's role in enhancing the autonomy of individual UxVs as well as the introduced challenges. Here, we extend this perspective to the swarm level, investigating how AI can enable collaborative behaviors and system-wide resilience. We firmly believe that AI can play a pivotal role in aspects such as adaptive communications, dynamic decision-making, and more robust security and self-adaptable mechanisms. Such advancements could potentially reduce or even eliminate the need for human controller supervision of each drone.

1.1. Our Contribution

Motivated by the observations outlined above, this review paper surveys SotA methodologies for implementing UxV swarms, while simultaneously encouraging further research across multiple facets of the field. Although there exists a wealth of reviews and analyses on UxVs—both as standalone platforms and as components within a swarm—many tend to either concentrate on narrowly defined technologies or provide only a high-level perspective on the evolution of drone and swarm systems.

In this context, the primary contributions of this paper are as follows:

- Present a thorough overview of the existing literature, i.e., reviews, surveys, etc., in the domain of swarm technologies and identify research gaps.
- Enhance understanding on how AI can be integrated in different swarm operations, e.g., path planning, formation control, etc.
- Investigate how AI can improve the swarm's networking, e.g., adaptive communication protocols, channel selection, etc.
- Enhance the security posture of the swarm with anomaly detection, self-recovery, and self-adaptability to external and internal factors.
- Analyze trends in vehicle selection to highlight dominant platforms and neglected domains.
- Discuss our findings and elaborate on future areas of research.

The novelty of our work lies in its agnostic and holistic approach to swarm intelligence in autonomous systems. Unlike existing reviews, which often focus on specific vehicle types, particular AI algorithms, or narrow application domains, our study deliberately avoids these constraints. We treat the swarm as a conceptual and functional entity. We analyze its behavior, coordination mechanisms, and emergent properties independent of the underlying hardware or software implementation.

1.2. Methodology

In this chapter, we describe the methodology used to collect and select the literature analyzed in this survey. The methodology we followed is compliant with the PRISMA 2020 guidelines. Our goal was to identify peer-reviewed studies on UxV swarms that apply AI methods to swarm intelligence, communication, and security. This ensures that our review is both comprehensive and reproducible. We adopted an iterative, keyword-

driven search strategy and queried the following electronic databases between April and June 2025:

- MDPI;
- IEEE Xplore Digital Library;
- SpringerLink;
- ScienceDirect;
- Scopus;
- Google Scholar.

To uncover research spanning all aspects of swarm intelligence, communication, and security, we searched using combinations of relevant keywords, including the following: “UxV swarm”, “UAV swarm”, “UGV swarm”, “UUV swarm”, “USV swarm”, “multi-robot swarm”, “swarm intelligence”, “AI in UxV swarms”, “machine learning in UxV swarms”, “reinforcement learning in UxV swarms”, “federated learning in UxV swarms”, “adaptive swarm communications”, “machine learning-driven swarm communications”, “swarm security”, “AI-driven swarm security”, “machine learning for swarm intrusion detection”.

Our results were filtered to include only peer-reviewed journal and conference papers. Citation trails of key publications were also followed to capture additional influential works that might not have been indexed under the same terms. We included studies that

- Focused on UxV swarms (UAV, UGV, USV, UUV, or mixed swarms);
- Applied AI-, ML-, or data-driven methods to swarm intelligence, communication, or security;
- Were published in English and peer-reviewed (journal or conference).

We excluded

- Non-peer-reviewed works such as patents, theses, and white papers;
- Duplicate records across databases.

All records were imported into a reference manager (Zotero), where duplicates were automatically and manually removed by matching DOI, title, and author. The authors did not use any automation tools to filter out possibly irrelevant records. Two authors independently screened titles and abstracts for relevance, resolving disagreements by consensus. The full text of the remaining titles was then retrieved by the authors. Subsequently, full-text screening was conducted to confirm that each included study was relevant to the thematic focus of our review. A summary of the described methodology, along with the counts of all the reports we collected and filtered, adapted from the PRISMA 2020 guidelines, is shown in Figure 1.

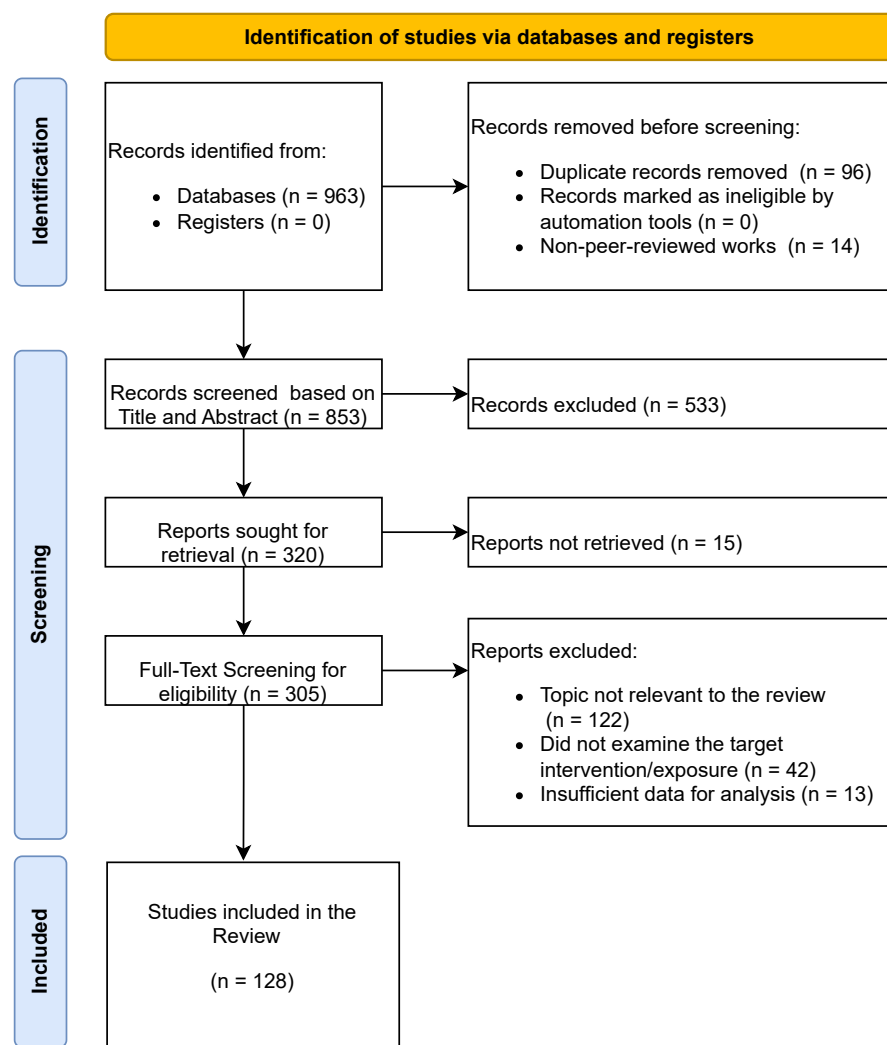


Figure 1. PRISMA 2020 flow diagram illustrating the literature identification, screening, eligibility, and inclusion process.

2. Previous Review and Survey Works

Previously, a wide range of reviews and surveys have been published investigating SI, swarm communications, and swarm security. Our goal in this chapter is to show whether there is relevant research that focuses on AI as the main technological enabler for the above sectors. The AI umbrella covers a plethora of algorithms, from rule-based ones to the more dynamic ones, e.g., neural networks (NNs) and machine learning (ML). In this review, we will focus on modern algorithms that handle ML, NNs, deep learning (DL), reinforcement learning (RL), deep reinforcement learning (DRL), and even large language models (LLMs). Consequently, we will examine whether the reviewed articles from Tables 1 and 2 address the following key areas, which are also presented in Figure 2:

- **AI in Swarm Intelligence:** Applications of AI to swarm coordination, path planning, resource allocation, target tracking, formation control, and other collective swarm behaviors.
- **AI-Based Adaptive Communications:** Use of AI to enable dynamic and adaptive communication systems, including adaptive routing protocols, frequency allocation, channel selection, topology control, and quality-of-service (QoS) optimization.
- **AI-Driven Security Mechanisms:** Techniques leveraging AI to enhance the security and resilience of the swarm and individual agents, including anomaly detection, intrusion detection, self-healing, fault tolerance, and automated incident response.

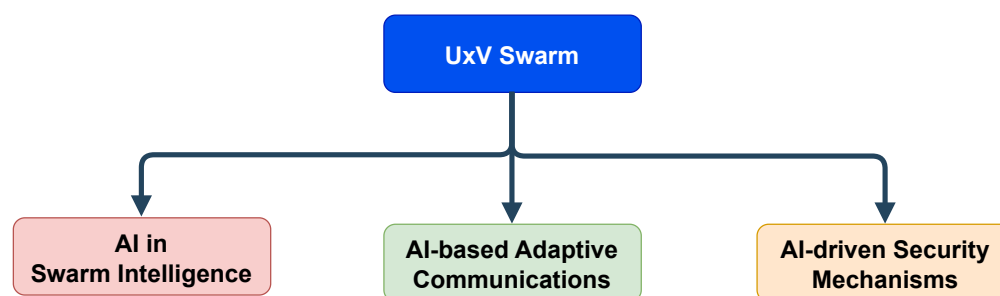


Figure 2. Key areas of interest in UxV swarms.

Tables 1 and 2 provide a high-level analysis of the collected surveys, using three types of annotations to indicate the extent of coverage across the key areas:

- ✓: Indicates that the survey offers a comprehensive analysis of the subject, supported by examples and in-depth discussion.
- ✗: Indicates that the survey does not address the subject at all.
- Partially: Indicates that the survey mentions the subject briefly or superficially, without detailed discussion or illustrative examples.

Table 1. Relevant review and survey papers.

Reference	Year	Short Description	AI in Swarm Intelligence	AI-Based Adaptive Communications	AI-Driven Security Mechanisms
Chen, Xi et al. [26]	2020	Centered on communication architectures and routing protocols, with only indirect references to optimization.	✗	✓	✗
Chen, Wu et al. [27]	2020	Proposes robustness techniques for drone swarms using rule-based models with limited AI coverage.	Partially	✗	✗
Sarkar et al. [28]	2023	Examines AI methods for network control and UAV coordination, with swarming treated as one of several enabling technologies.	✓	✓	Partially
Puente-Castro et al. [29]	2022	Focused survey on AI-driven path planning for UAV swarms using RL, SI, and GNN, without discussing swarm security aspects.	✓	✗	✗
Abdelkader et al. [30]	2021	Discusses UAV swarms through the lens of localization and planning, minimizing AI and communication complexity.	Partially	Partially	✗
Tahir et al. [31]	2019	Engineering-oriented survey of UAV swarms focusing on control systems, decentralized coordination, and layered communications.	✗	✗	✗

Table 1. Cont.

Reference	Year	Short Description	AI in Swarm Intelligence	AI-Based Adaptive Communications	AI-Driven Security Mechanisms
Li et al. [32]	2024	Integrates AI for adaptive control in heterogeneous UxV networks, centered around SDN architecture rather than swarm systems.	Partially	✓	✗
Schranz et al. [33]	2020	Provides a taxonomy of swarm behaviors with examples, avoiding technical AI or communication system analysis.	✗	Partially	✗
Alqudsi et al. [34]	2025	Holistic review integrating AI, ML, communication networks, and ethics in design of scalable, intelligent UAV swarms.	✓	Partially	✗
Zaitseva et al. [35]	2023	Examines reliability modeling and structural analysis of drone swarms, treating swarms as topological systems, not AI-driven.	✗	✗	✗
Asaamoning et al. [36]	2021	Reviews drone swarms as distributed control systems, highlighting self-organizing algorithms, system security, and resource management.	✓	Partially	✗
Chung et al. [37]	2018	Comprehensive review of UAV swarms, covering flight dynamics, control algorithms, and 3D autonomy.	Partially	✗	✗
Connor et al. [38]	2021	Reviews underwater swarm platforms, sensor types, and role-based designs with minimal focus on AI or SI.	Partially	✗	✗

Table 2. Relevant review and survey papers.

Reference	Year	Short Description	AI in Swarm Intelligence	AI-Based Adaptive Communications	AI-Driven Security Mechanisms
Wu et al. [39]	2022	Research on collaborative search and hunting using multiple USVs. AI central in functions such as path planning, task allocation, and collaborative search.	✓	✗	✗
Vedachalam et al. [40]	2019	Discusses AUV design and reliability for strategic use, emphasizing battery modeling and navigation accuracy.	Partially	✗	✗

Table 2. *Cont.*

Reference	Year	Short Description	AI in Swarm Intelligence	AI-Based Adaptive Communications	AI-Driven Security Mechanisms
Liu et al. [41]	2023	Focuses on coordination challenges of underwater swarms, highlighting environmental constraints and communication limitations.	✗	Partially	✗
Boccardo et al. [42]	2021	Summarizes real-world use cases and scenarios and cites works that implemented/tested UAV systems in practice.	Partially	Partially	Partially
Michailidis et al. [43]	2022	Examines the possibility of a UAV-aided IoT network. Focuses on networking and secure communications.	Partially	✓	✓
Zhou et al. [44]	2020	Examines SI by focusing on decision-making, path planning, control, communication, and applications.	✓	Partially	✗
Javed et al. [45]	2024	Covers aspects of UAV swarms such as formation control, path planning, autonomy, coordination, security, and communications.	✓	✓	✓
Liu et al. [46]	2024	Aims to provide a comprehensive survey on applications, challenges, and opportunities of GAI in unmanned vehicle swarms.	✓	✓	✓
Azoulay et al. [47]	2021	Offers a rich and methodical review of ML techniques for UAV flock formation, task allocation, and coordination	✓	Partially	✗

2.1. AI in Swarm Intelligence

Research into AI-driven swarm intelligence (SI) is diverse, encompassing both rule-based approaches and advanced AI methodologies. We need to clarify here that we adopt the definition of SI from [48], which defines it as a mindset of intelligent groups of simple agents that exchange information with each other or with the environment following basic rules that collectively lead to global system behavior. Sarkar et al. [28] place AI as the central enabler for swarm coordination. They leverage ML, DL, RL, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) for tasks such as trajectory optimization and resource allocation. Their approach aligns closely with our focus on AI as a foundational swarm technology, although they only partially cover security aspects. Alqudsi et al. [34] similarly adopt AI for autonomy, obstacle avoidance, and decision-making, but omit security mechanisms. In a similar manner, the researchers in [44] place AI as the cornerstone of SI, by highlighting the role of ML, DL, and RL and evolutionary algorithms in decision-making, autonomous control, path planning, and environment perception. Following the same trend, Javed et al. similarly adopt AI (RL, NNs, clustering, SVMs) for task coordination, path planning, formation control, autonomous decision-

making, and data processing in [45]. The work in [46] stands out by choosing generative artificial intelligence (GAI) as the main technological enabler in tasks like path planning, state estimation, and trajectory modeling. Finally, in [47], Azoulay et al. try to apply ML methods to various swarm control and flocking behaviors and cover subjects such as flock formation and clustering, task allocation, multi-agent coordination, trajectory optimization, and navigation.

Chen, Wu et al. [27] explore SI within flying ad hoc networks (FANETs), applying percolation theory to maintain network topology using only local information. While the work addresses dynamic topology control, it remains primarily rule-based, not employing ML or NNs. Asaamoning et al. [36] apply SI algorithms such as Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Raft consensus protocols to path planning and formation control. Although they acknowledge AI-enhanced routing, their approach is algorithmic, not data-driven. Schranz et al. [33] prefer bio-inspired algorithms to AI, focusing on decentralized communication between swarm members. Wu et al. [39] similarly emphasize heuristic methods without delving into AI.

In summary, while AI-based approaches to SI are gaining significant traction, much of the existing research still relies on rule-based or bio-inspired algorithms, indicating that there is still a gap for more data-driven AI methodologies in swarm coordination.

2.2. AI-Based Adaptive Communications

In the context of adaptive communications, Li et al. [32] implement DRL for power control in software-defined UxV networks, aligning well with our focus on AI-based communications. They also use unsupervised learning for drone controller placement but only mention security briefly as a future direction. Michailidis et al. [43] focus on the intersection of UAVs, internet of things (IoT), and mobile edge computing (MEC), emphasizing AI-driven communication optimization. They explore the role of AI—especially federated learning (FL), RL, and DL—in dynamic task offloading, adaptive communication, and secure data management. We also witness a shift toward AI-based approaches in [45], where researchers examine ML-based topology prediction and RL-based adaptive routing protocols. Very interesting research is presented in [46], where issues like data restoration, bit error rate improvements, and spatio-temporal user distribution prediction are addressed by GAI algorithms, e.g., a conditional generative adversarial network (GAN), diffusion models, and transformers.

Chen, Xi et al. [26] provide a comprehensive overview of communication architectures, including UAV-to-UAV and UAV-to-infrastructure networks. They discuss ring, star, and mesh topologies but do not incorporate AI-driven adaptation mechanisms. Similarly, Vedachalam et al. [40] thoroughly explore communication systems (acoustic, optical) but exclude AI-driven optimization, focusing instead on engineering challenges and redundancy. Liu et al. [41] similarly limit their communication analysis to underwater vehicles, excluding AI-driven protocols. Boccadoro et al. [42] provide a technical survey on the internet of drones (IoD), covering inter-agent communication and network configurations comprehensively. While they acknowledge AI as a potential tool for optimization, specific implementations are not addressed.

Overall, the literature indicates that while communication architectures are well explored, the integration of AI-driven adaptive communication techniques remains underdeveloped, highlighting a potential research direction in real-time, AI-based communication strategies.

2.3. AI-Driven Security Mechanisms

Sarkar et al. [28] touch on security through AI-enabled anomaly detection, focusing mainly on identifying malicious nodes or patterns of erratic behavior that could disrupt UAV networks. While their approach demonstrates AI's potential for anomaly detection, its scope is limited to specific attack scenarios without proposing comprehensive frameworks. Zaitseva et al. [35] adopt a structural reliability perspective, assessing vulnerabilities in communication links as potential points of failure in the swarm. Although AI is not directly implemented, their work underscores opportunities for AI-driven fault tolerance mechanisms. Asaamoning et al. [36] identify security as a challenging area in networked control systems, suggesting AI-enhanced routing as a potential future direction, but not integrating it in practice. Boccadoro et al. [42] address cryptographic challenges and UAV-specific threats, proposing AI as a tool for intrusion detection, although the concept remains largely theoretical. Chung et al. [37] emphasize control-theoretic approaches to security, focusing on fault detection through optimized path planning. AI applications are not discussed, but their framework offers potential integration points for anomaly detection systems.

The most complete piece of work we have witnessed so far covering the intersection between security and AI is [43]. Michailidis et al. stand out for their in-depth treatment of AI-driven security in UAVs and IoT environments. They explore the use of FL, RL, and DL for intrusion detection, adaptive communication optimization, and blockchain-based authentication. Javed et al. consider blockchain as a possible security framework in [45], but they also elaborate on ML for detecting intrusion and spoofing attempts as well as GPS manipulation. Moving one step further, Liu et al. [46] introduce LLMs and transformers in intrusion detection methods. They even examine the use of GANs for trajectory obfuscation and image privacy preservation.

Still, while we observe a trend toward the adoption of AI as a security enabler, practical implementations of AI-driven security mechanisms in swarm systems remain scarce, suggesting a significant research gap in developing real-time, AI-based anomaly detection and mitigation frameworks.

2.4. Comparative Analysis

In terms of our analysis of the literature, we believe that our work stands out from existing surveys, although some studies adopt a similar approach and methodology. The closest are [45,46], yet there are important differences. Specifically, ref. [46] focuses primarily on GAI and its future potential in swarms, while presenting only limited practical applications. Ref. [45], on the other hand, is the most similar to our work but restricts its scope to UAVs, excluding other vehicle types. This is also evident in [39]. Such approaches narrow the scope of research by focusing on one type of vehicle and its intricacies. Furthermore, we believe that the majority of the literature lacks a holistic approach and focuses on a specific aspect of swarm systems without integrating others. For example, in [28], researchers address SI and communication, but are not concerned about how these functionalities remain secure. Other works present the swarm only as a means of improving other systems, such as in [43], where it is used to improve IoT networks. Our work, on the contrary, provides a comprehensive perspective that brings together different vehicle classes while examining how AI can enhance swarm-level autonomy, communication, coordination, and security in an integrated manner.

3. AI in Swarm Intelligence

In this section, we will investigate how AI can contribute to and enhance SI and swarm operations in general. Our focus is to identify the swarm operations that are most distinct

from swarm logic and to describe how researchers address them, primarily using AI and ML techniques. Based on pieces of work such as [49,50], we have identified the following core operations, which we also present in Figure 3:

- **Path Planning and Navigation:** These involve how both individual UxVs and the whole swarm plan their routes within the operating environment.
- **Obstacle and Collision Avoidance:** This concerns how UxVs and the swarm avoid environmental obstacles and internal collisions.
- **Formation Control:** This describes how UxVs form and maintain formations, as well as how they adapt these formations dynamically.
- **Target Search:** This includes detecting targets, pursuing them efficiently, and ultimately reaching them.
- **Mission Planning:** This addresses how the swarm can operate more effectively given specific goals and environmental conditions.
- **Task Optimization:** This focuses on how the swarm efficiently performs assigned tasks and allocates them to individual UxVs.
- **Resource Optimization:** This primarily relates to energy efficiency and how the swarm can complete tasks without depleting its resources.

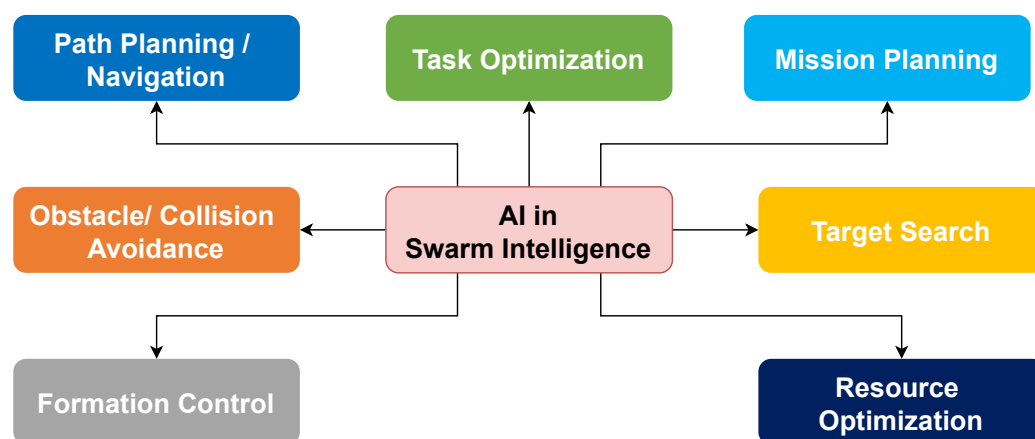


Figure 3. AI in swarm intelligence.

Since our survey examines AI as a technological enabler, we also need to examine the algorithms and methodologies covered by each paper. For this reason and since many papers covered multiple swarm operations, we have chosen to classify them based on which technology they relied on the most. This is why we have chosen to split this chapter into three sub-chapters:

- **Approaches Relying on Reinforcement Learning:** This chapter mostly includes academic papers which rely on RL and algorithms from the RL family to perform the swarm operations.
- **Supervised, Federated, and Probabilistic Models:** In this section, we deal with pieces of work which rely on supervised NNs, FL, or probabilistic approaches.
- **Hybrid, Bio-Inspired, and Other Methods:** Finally, this category includes research that relies on bio-inspired algorithms, e.g., ACO, PPO, etc., and approaches that do not belong in any of the previous categories.

3.1. Approaches Relying on Reinforcement Learning

In Table 3, we have gathered the relevant literature and marked which swarm operations are included in each piece of the literature. These works in the literature are highly reliant on RL and DRL. For UAV swarms, Xiao et al. [51] introduce a curriculum-based mul-

tistage RL framework and perform visual target search using RGB input, while Perrusquia et al. rely on observational trajectory data in [52] and introduce a multi-agent inverse RL framework to determine the mission rewards and network topology of the swarms. Both approaches achieve decentralized coordination since agents rely on raw sensor input. For hybrid UAV–UGV scenarios, researchers in [53] address formation control, mission planning, and navigation of hybrid drone swarms by training a UAV to shepherd UGVs using a deep hierarchical RL approach. The shepherding task is decomposed into two subtasks, collect and drive, and two double deep Q-networks (DQNs) are trained for each task. Zhang et al. also tackle mission planning, task optimization, and path planning for heterogeneous UGV-UAV coalitions in [54] by introducing augmented DRL, a hybrid framework that integrates generative adversarial imitation learning with proximal policy optimization (PPO) to learn complementary cooperative strategies.

Other studies emphasize robustness under uncertainty. In [55], Venturini et al. develop a double deep Q learning (DDQL)-based distributed RL framework for UAV swarms to perform efficient target search under partial observability and possible communication loss. Qamar et al. extend DRL with island policy-based optimization in [56]. This technique splits the UAV swarm into sub-swarms, which perform tasks such as formation control, navigation, dynamic obstacle avoidance, and multi-target tracking in 3D environments. Staying on the USV use case, Gan et al. [57] propose a cooperative target-chasing and encirclement framework for multiple USVs. With DRL at its basis, USV agents are able to execute encirclement maneuvers and avoid collisions. In [58], Cao and Fang deal with path planning and collision avoidance for multiple UGVs in static environments by proposing the Optimized-Weighted-Speedy Q-learning algorithm, a method that integrates a distributed anti-collision cooperation mechanism that enables agents to simultaneously compute collision-free paths. Finally, the researchers in [59] tackle issues like target search, formation control, and mission planning by proposing a dynamic switching-enabled multi-agent RL (MARL) framework, which enables autonomous UUV swarms to track multiple targets cooperatively while adapting to sonar noise and ocean currents.

Table 3. Studies with approaches relying on reinforcement learning.

Reference	Vehicle	Path Planning and Navigation	Obstacle and Collision Avoidance	Formation Control	Target Search	Mission Planning	Task Optimization	Resource Optimization
Xiao et al. [51]	UAV	✓	✓		✓			
Perrusquia et al. [52]	UAV					✓	✓	✓
Nguyen et al. [53]	UAV, UGV	✓		✓	✓			
Zhang et al. [54]	UAV, UGV	✓	✓			✓	✓	
Venturini et al. [55]	UAV	✓	✓	✓	✓			✓
Qamar et al. [56]	UAV	✓	✓	✓	✓	✓		✓
Gan et al. [57]	USV	✓	✓	✓	✓	✓	✓	✓

Table 3. Cont.

Reference	Vehicle	Path Planning and Navigation	Obstacle and Collision Avoidance	Formation Control	Target Search	Mission Planning	Task Optimization	Resource Optimization
Cao et al. [58]	UGV	✓	✓	✓				
Wang et al. [59]	UUV	✓		✓	✓	✓		

Although these works collectively demonstrate the adaptability of RL to various swarm tasks, we should mention possible limitations. First, we need to address the fact that most RL frameworks rely on simulation environments for training, raising concerns about transferability to real-world deployments where sensing is noisy and dynamics are uncertain. Furthermore, we should also raise the question of how an RL framework adapts when swarm size increases, which could introduce unexpected behaviors. Efforts such as distributed or hierarchical RL, e.g., refs. [53,55], partially address this by decomposing tasks or distributing policies, but these solutions introduce added complexity in synchronization and convergence.

3.2. Supervised, Federated, and Probabilistic Models

Apart from RL and already established bio-inspired methods, other types of ML models and hybrid approaches, combining AI and probabilistic techniques, have been proposed. In Table 4, we have gathered the relevant literature with the corresponding swarm operations. In [60], Yuwen et al. tackle formation control, mission planning, and task optimization in USV swarm confrontations by proposing a distributed Nash equilibrium seeking algorithm within a coalition game framework for uncertain Euler–Lagrange systems with local and coupling constraints. The researchers in [61] also use probabilistic models and present a model-driven cooperative path planning strategy for dynamic target search in multi-UUV formations, where spiral path parameterization, hierarchical formation control, and Kriging-assisted discrete global optimization are used to maximize the target detection success rate under dynamic and uncertain conditions. Paczyk et al. [62] present a neural control system for autonomous UUV swarms using RNNs, a type of NN not so famous in this field of research, for formation control and an autoencoder-based filter for sonar noise in obstacle avoidance.

Raj et al. [63] introduce a feed-forward NN-based mechanism for UAV trajectory prediction and collision avoidance, incorporating various activation functions, including a novel AdaptoSwelliGauss function. Cabuk et al. [64] focus on energy optimization within UAV swarms, training four predictive energy models on real-world flight data and deriving a cost function to evaluate swarm-wide energy consumption for operations such as connectivity restoration, enabling energy-aware reconfiguration. Kussyk et al. [65] propose a decentralized AI-based flight control system where individual UAVs compute their next positions using a genetic algorithm and communicate only with nearby neighbors, enhancing swarm efficiency and scalability without centralized control. Zeng et al. [66] present a federated learning framework in which each UAV trains a local model and sends it to a leading UAV for aggregation into a global model, which is then redistributed to the swarm. Finally, Trihinas et al. [67] introduce a framework for repeatable testing of ML-driven UAV applications, supporting mission workflow prototyping, energy profiling, and performance monitoring of both tasks and vehicles.

These supervised, federated, and probabilistic approaches highlight important trade-offs compared to RL. Probabilistic methods such as those in [61] offer interpretability and robustness under uncertainty, but their reliance on handcrafted models and assumptions about dynamics may limit generalization to heterogeneous swarms or unstructured environments. Supervised learning approaches, e.g., refs. [63,64], benefit from efficiency and use of real-world data, yet they often require labeled datasets and may lack adaptability to unforeseen scenarios. Federated learning frameworks address scalability by distributing training across agents, but they introduce challenges such as model heterogeneity, synchronization, and communication costs.

Table 4. Studies with approaches relying on supervised, federated, and probabilistic Models.

Reference	Vehicle	Path Planning and Navigation	Obstacle and Collision Avoidance	Formation Control	Target Search	Mission Planning	Task Optimization	Resource Optimization
Yuwen et al. [60]	USV			✓		✓	✓	
Qin et al. [61]	UUV	✓		✓	✓			
Praczyk et al. [62]	UUV	✓	✓	✓				
Raj et al. [63]	UAV	✓	✓					
Cabuk et al. [64]	UAV						✓	✓
Kusyk et al. [65]	UAV	✓	✓	✓				✓
Zeng et al. [66]	UAV						✓	✓
Trihinas et al. [67]	UAV					✓	✓	✓

3.3. Hybrid, Bio-Inspired, and Other Methods

SI algorithms and bio-inspired optimization techniques, like ACO, PSO, and Artificial Bee Colony (ABC), offer fertile ground for enhancement through ML or other techniques, making them more dynamic and adaptable to stochastic environments. Table 5 includes the pieces of work that rely on such methods to execute the swarm operations. For example, in [68], Saeed et al. analyze how swarm intelligence algorithms, i.e., ACO, PSO, and ABC, can be enhanced for UAV path planning in 2D and 3D environments. They integrate obstacle distance into ACO's pheromone update rules, which significantly optimizes trajectory planning and obstacle avoidance. Gal goes one step further in a use case with USV swarms in [69], by proposing a multi-target tracking approach based on adaptive PSO with K-Nearest Neighbors (KNN), enhanced with a pursuit–evasion model. Similarly, in [70], the researchers perform target search for UAV swarms, using AI-driven strategies, such as Lawn Mower, Hill Climbing, and A-star with Bayesian updating. Zeng and Nait-Abdesselam propose an RL-enhanced Boid model for UAV swarms in [71]. This approach enables the swarm agents to learn flocking behavior through multi-agent Q-learning, offering robust decentralized control for swarm navigation and cohesion, surpassing rule-based approaches. In [72] the researchers again introduce an AI ground station called the Hive, which largely decentralizes the swarm and enables autonomous

reconnaissance missions. Distributed PSO alongside vision-based navigation is used to enhance formation control, task execution, and long-term mission planning.

Zhu et al. focus on optimizing tasks and resources in [73]. They develop a mathematical model for multi-USV maritime patrol task assignment under endurance and time constraints. By integrating A* path planning and energy-aware scheduling, the system ensures efficient patrol task coverage with minimized deviation from task time windows and optimized energy resupply. Liu et al. [74] propose a hierarchical mission planning system for multi-UGV swarms that integrates task decomposition, assignment, sequencing, and path planning using a hybrid clustering method and a multilayer planner. Key techniques include A-star, ACO, and a novel multi-operator continuous ACO. In [75], Hussein et al. investigate ML-based shepherding algorithms to control swarms of UxVs in complex environments. They rely on human demonstrations to train ML controllers and explore how different levels of autonomy during data collection impact swarm control performance. Finally, in [76], Saffre et al. propose a bio-inspired, decentralized framework for UAV swarms performing long-term surveillance missions. The system relies on a shared digital twin that models environmental signals to guide individual drone path planning and task allocation.

SI and bio-inspired approaches remain attractive for swarm coordination due to their simplicity, scalability, and ability to exploit emergent behaviors. When combined with ML, they can adapt to dynamic conditions and outperform purely rule-based methods, as in [71]. However, their reliance on heuristics or handcrafted mechanisms often limits generalization across domains. For example, adaptations such as modified pheromone rules or pursuit–evasion models improve task-specific performance but may not scale efficiently to heterogeneous swarms or large agent populations. Similarly, optimization-driven frameworks, e.g., refs. [73,74], provide strong guarantees on task allocation and resource use but may incur computational complexity that hinders real-time deployment.

Table 5. Studies with approaches relying on hybrid, bio-inspired, and other approaches.

Reference	Vehicle	Path Planning and Navigation	Obstacle and Collision Avoidance	Formation Control	Target Search	Mission Planning	Task Optimization	Resource Optimization
Saeed et al. [68]	UAV	✓	✓					✓
Gal [69]	USV	✓			✓	✓	✓	✓
Sundelius et al. [70]	UAV	✓			✓	✓	✓	
Zeng and Nait-Abdeslam [71]	UAV	✓	✓	✓			✓	
Awasthi et al. [72]	UAV	✓		✓		✓		✓
Zhu et al. [73]	USV						✓	✓
Liu, Jing et al. [74]	UGV	✓				✓	✓	
Hussein et al. [75]	Generic	✓	✓	✓				
Saffre et al. [76]	UAV				✓	✓		

3.4. Conclusion

Concluding our overview of AI in swarm operations, we would like to point out some interesting findings. Based on Tables 3–5, it is evident that aerial vehicles are dominant and much more preferable than others. This is expected, if we consider how modular and accessible commercial-off-the-shelf (COTS) equipment for UAVs is. Another explanation could be that UAVs operate in the sky without many obstacles, whereas UGVs or UUVs might encounter many times more obstacles. This significantly reduces the complexity of tasks such as path planning and formation control. In terms of swarm operations encountered in the bibliography, if we look at Figure 4, it has become pretty evident that path planning is the most well researched, with formation control following. We witnessed that formation control and collision avoidance overlapped in many pieces of work, which makes sense since formation control needs to take into account the possibility of collisions inside the swarm. Furthermore, for environmental obstacles, a visual way of identifying objects was preferred. Regarding task optimization, we need to state that while not being the main point many a time, a lot of papers considered and proposed ways to further optimize the core operations, like path planning or obstacle avoidance. Researchers had a similar approach to resource optimization, oftentimes presenting it as something complementary to more significant functionalities, e.g., path planning algorithms that take into account energy consumption.

Regarding the type of AI that was preferred in the sample of the literature, we observe that Tables 3–5 are of similar size. This distribution suggests that the swarm operations we identified have been repeatedly studied as technology has advanced, ranging from earlier methods such as ACO and PSO to more recent approaches based on RL, FL, etc. Note that paradigms with RL and FL at the center have been the most prevalent in recent years.

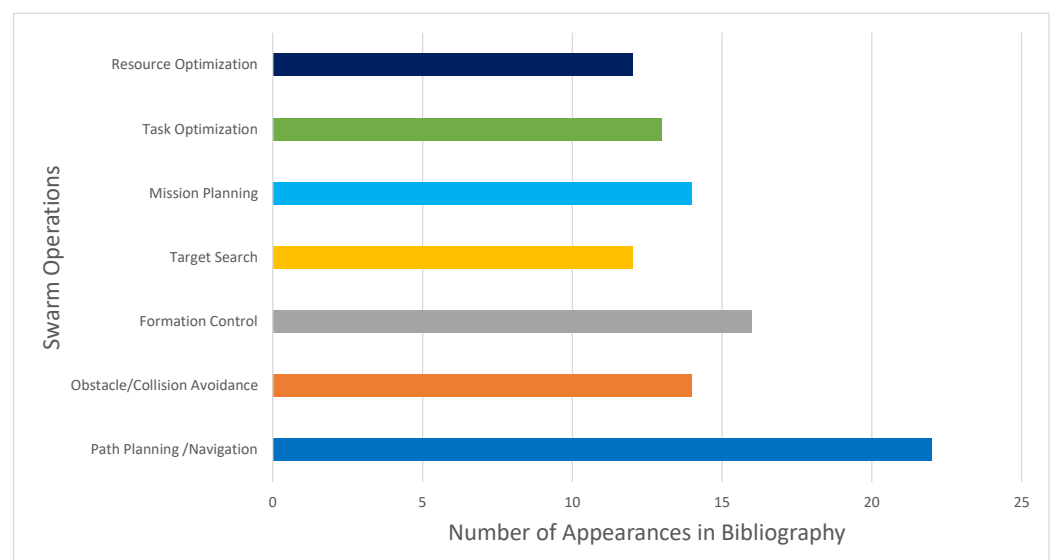


Figure 4. Appearances of each swarm operation in the surveyed literature.

4. AI-Based Adaptive Communications

In the previous chapter, we analyzed how AI can be integrated into swarm operations to make them more efficient and dynamic. In this chapter, we plan to cover the various communication aspects that connect individual UxVs and elevate them to the swarm level, as well as how AI can enhance these aspects. Following the same methodology as in the previous chapter, we will identify which communication aspects each paper addresses and

how AI is utilized. In Figure 5 we classify these communication functions in the following sub-categories:

- **Network Architecture and Topologies:** These address how nodes (UAVs, relays, base stations) are physically or logically arranged, and how coverage and connectivity are maintained—particularly in dynamic, multi-node environments.
- **Channel and Propagation Modeling:** This involves how signals behave during transmission, as well as methods to model, predict, measure, or enhance signal behavior.
- **Signal Quality:** This refers to signal processing techniques that improve signal fidelity, energy efficiency, or robustness.
- **Routing and Protocols:** These describe how data paths are selected and how nodes communicate using predefined or adaptive rules.
- **Network Management and Optimization:** These encompass system-level strategies for dynamically allocating resources, maintaining service quality, and optimizing overall network performance.

We will follow the same logic from Section 3 and again examine the literature we found based on the types of algorithms and methodologies they adopted:

- **Approaches Relying on Reinforcement Learning:** This chapter mostly includes academic papers which rely on RL and algorithms from the RL family to perform the swarm operations.
- **Supervised, Federated, and Probabilistic Learning Models:** In this section, we deal with pieces of work which rely on supervised NNs, FL, or probabilistic approaches.
- **Hybrid, Bio-Inspired, and Other Methods:** Finally, this category includes research that relies on bio-inspired algorithms, e.g., ACO, PPO, etc., and approaches that do not belong in any of the previous categories.

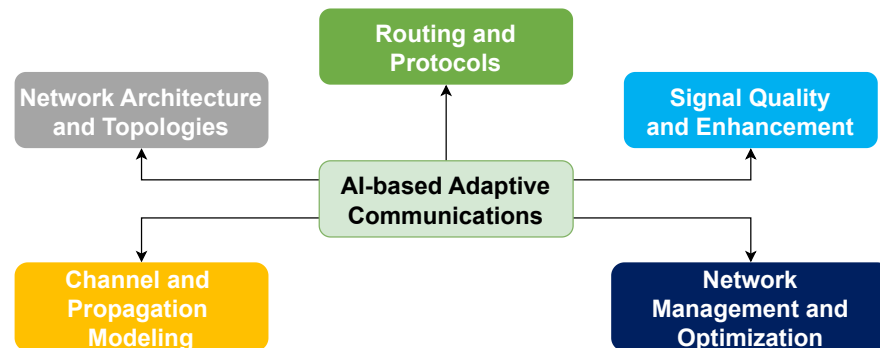


Figure 5. AI-based adaptive communications.

4.1. Approaches Relying on Reinforcement Learning

As in other swarm operations, RL and DRL still remain relevant in enhancing the swarm's communication posture. This is emphasized in surveys and reviews such as [77–79]. We present the relevant literature in Table 6 alongside the communication functions covered in said literature. Kuranthan et al. offer a comprehensive review in [77], elaborating on how ML-driven UAV swarms benefit from supervised, unsupervised, RL, and FL techniques in operations such as channel estimation, scheduling, trajectory planning, and resource management. In [78], Divakar et al. highlight how FL and RL, combined with SI, can optimize resource allocation, trajectory planning, and scalable communication in next-generation aerial networks. Similarly, in [79], Khan et al. advocate for RL, DRL, and FL in operations such as real-time path planning, dynamic routing, signal optimization, and distributed network intelligence. We observe that in [80], Kushik et al. present a DQN-based positioning system for UAV swarms, where UAVs act as relays to bridge broken radio frequency (RF) links. The system adapts to dynamic topology and varying

link conditions using parameters such as the signal-to-interference-plus-noise ratio (SINR), packet delivery ratio (PDR), and interference. Likewise, the researchers in [81] propose a DRL-powered integrated communication and control framework for autonomous UAV swarms. The leading UAV uses a deep deterministic policy gradient (DDPG) agent and optimizes control signal generation, subcarrier allocation, and data rate to manage formation while minimizing communication error and collision risks. Wang et al. propose an ML-powered framework for spectrum sharing in UAV swarm communication systems in [82]. They address challenges like jamming resistance, ultra-density, and dynamic topology through MARL, game theory, and task-driven optimization and achieve adaptive and collaborative spectrum usage.

Collectively, these studies underscore the promise of RL and its subdomains in enabling adaptive, resilient communication among swarms. However, significant challenges remain. The most influential factor is the diverse nature of communications, where different types of applications, e.g., video streaming or telemetry data, rely on different frequencies, protocols, and equipment. Video streaming, in particular, demands high bandwidth, stable bitrates, and low latency to ensure continuous quality of service (QoS). Creating a simulation environment that considers so many variables, which constantly change, is computationally demanding and could struggle with the sim-to-real gap, therefore significantly reducing the QoS for certain applications. Furthermore, while the spectrum sharing and relay positioning frameworks, e.g., refs. [80,82], show robustness to jamming and topology changes, such algorithms should be tested by adding or removing swarm agents constantly, making agents lose connection temporarily, or using more powerful jammers if possible.

Table 6. Studies with approaches relying on reinforcement learning.

Reference	Vehicle	Network Architecture and Topologies	Channel and Propagation Modeling	Signal Quality and Enhancement	Routing and Protocols	Network Management and Optimization
Kurunathan et al. [77]	UAV	✓	✓		✓	✓
Divakar et al. [78]	UAV	✓		✓	✓	
Khan et al. [79]	UAV		✓	✓	✓	✓
Koushik et al. [80]	UAV	✓			✓	✓
Wei et al. [81]	UAV		✓	✓	✓	✓
Wang et al. [82]	UAV				✓	✓

4.2. Supervised, Federated, and Probabilistic Models

With works such as [77] in mind, we move from RL approaches to more supervised and probabilistic solutions. Table 7 includes the scientific publications we have discovered on this matter accompanied by the communication function each publication covers. In [83], Mao et al. introduce a DL-based transport layer protocol for UAV swarms. Their approach uses network coding and real-time topology-aware control to distinguish between congestion and random loss and also predicts optimal coding rates based on dynamic link conditions. The researchers in [84] propose a decentralized FL framework for UAV swarms using the inexact stochastic parallel random walk alternating direction method of multipliers algorithm, enabling efficient model training under dynamic topologies with low communication overhead. Xu et al. rely again on FL in [85] and present a hierarchical FL-based semantic communication framework for UAV swarms. Each UAV trains its semantic model according to local data; all model parameters are then aggregated to achieve global parameter consensus. Finally, the researchers in [86] introduce a scalable drone

swarm routing system that combines multi-beam beamforming with a feed-forward NN to optimize signal fidelity and path planning. By learning the deviation between model-based routing outputs and ground-truth signals, the system passively supports decentralized and real-time routing in dense UAV networks.

Moving to more probabilistic approaches, the researchers in [87] introduce an ML-guided UAV deployment framework for emergency cellular networks. They use k-medoids clustering to optimally place aerial base stations to enhance user signal quality. Their approach improved coverage and SINR resilience even under localization errors, offering a robust solution for rapid network restoration in disaster scenarios. Comparably, Khalil et al. propose an ML-driven UAV swarm communication model for search-and-rescue missions in [88]. They leverage random forest (RF) regression for accurate channel modeling and K-means clustering for adaptive swarm formation in order to ensure intra-swarm link quality prediction and efficient swarm management. Khalil et al. performed similar research in [89]. In their paper, they present an ML framework that predicts and mitigates outage probabilities in 6G UAV swarm relays by combining RF regression for attenuation forecasting, K-means clustering for band selection, and custom metasurface design for signal enhancement. In [90], Lim et al. introduce a Bayesian network classifier and predict the reliability of the communication of UAV swarms by incorporating latency constraints and identifying the likely failure modes.

These supervised, federated, and probabilistic approaches offer complementary advantages compared to RL. Supervised and DL-based frameworks such as those described in [83,86] are efficient at learning from structured data and optimizing specific communication functions, but they require large, labeled datasets and may struggle to generalize under unpredictable conditions. Federated learning, e.g., refs. [84,85], enhances scalability and reduces central bottlenecks, yet it introduces synchronization challenges and communication overhead that may offset its benefits in bandwidth-constrained aerial networks. Probabilistic and statistical models provide interpretability and robustness under uncertainty, making them well suited for emergency or highly dynamic scenarios. However, they often rely on simplifying assumptions and may face computational limits in real-time large-scale deployments.

Table 7. Studies with approaches relying on supervised, federated, and probabilistic Models.

Reference	Vehicle	Network Architecture and Topologies	Channel and Propagation Modeling	Signal Quality and Enhancement	Routing and Protocols	Network Management and Optimization
Kurunathan et al. [77]	UAV	✓	✓		✓	✓
Mao et al. [83]	UAV		✓	✓	✓	
Xiao et al. [84]	UAV	✓				✓
Xu et al. [85]	UAV		✓	✓		
Myers et al. [86]	UAV		✓	✓	✓	✓
Tsipi et al. [87]	UAV	✓		✓		✓
Khalil et al. [88]	UAV		✓	✓		✓
Khalil et al. [89]	UAV	✓	✓	✓		
Lim et al. [90]	UAV	✓	✓		✓	

4.3. Hybrid, Bio-Inspired, and Other Methods

The approaches presented in this section combine biologically inspired coordination and signal-level innovations to address the communication aspects of the swarm. The

literature that exploits said approaches is presented in Table 8. In [91], the researchers demonstrate the strong potential of massive multiple-input and multiple-output (MIMO) systems by providing closed-form ergodic rate bounds, incorporating 3D geometry, polarization mismatch, and UAV mobility to optimize antenna array configurations and ensure reliable high-capacity links. Lakas et al. preserve swarm cohesion in [92] by proposing a leader–follower-based multi-cluster swarm control system using a simple 1-hop broadcast protocol for coordination and connectivity maintenance. On the other hand, Sousa et al. rely on multi-hop communication in [93] and present a system for aquatic USV swarms using a passive link quality estimation strategy to optimize data routing. Their system incorporates three acknowledgment-based delay-tolerant network protocols, validated in both real and simulated environments. Similarly and in order to increase network quality, Nomikos et al. propose a buffer-aided opportunistic UAV swarm framework for 6G maritime networks in [94], combining non-orthogonal multiple access (NOMA) and dynamic decoding to boost sum-rate and reduce delay under poor shore-to-sea link conditions.

Ultimately, we will examine a number of hybrid approaches that integrate both ML and bio-inspired algorithms. A good example of such a hybrid approach can be found in [95]. Pham et al. tackle sum-rate maximization for UAV-assisted visible light communication (VLC) with NOMA by combining swarm intelligence (Harris Hawks optimization) and an NN for joint UAV placement and power allocation. In [96], Zhou et al. propose a hybrid edge computing framework for the IoD, combining DL for network load prediction with swarm-based routing for dynamic path adaptation. This integration enables responsive, low-latency communication and efficient resource management in IoD environments. Finally, the researchers in [97] propose a joint SI and DL framework to optimize UAV relay positioning, hybrid beamforming, and power allocation in UAV-assisted mmWave massive MIMO IoT networks. The solution enhances throughput and reduces delay with a low-complexity NN capable of near-optimal performance.

Bio-inspired coordination and signal-level innovations demonstrate strong potential for enhancing swarm communication, particularly in challenging environments such as maritime or ultra-dense UAV networks. Signal-level methods, such as MIMO and NOMA, provide high-capacity, low-latency links but are highly sensitive to mobility, interference, and hardware constraints. Approaches such as leader–follower and multi-hop are lightweight and scalable, yet may struggle with dynamic link quality and unpredictable disruptions. Hybrid frameworks that integrate SI with ML, e.g., refs. [95–97], attempt to balance robustness and adaptability, but they introduce additional complexity in terms of computation and synchronization.

Table 8. Studies with approaches relying on hybrid, bio-inspired, and other Methods.

Reference	Vehicle	Network Architecture and Topologies	Channel and Propagation Modeling	Signal Quality and Enhancement	Routing and Protocols	Network Management and Optimization
Chandhar et al. [91]	UAV		✓	✓		
Lakas et al. [92]	UAV	✓				✓
Sousa et al. [93]	USV	✓			✓	
Nomikos et al. [94]	UAV	✓	✓			✓
Pham et al. [95]	UAV		✓	✓	✓	✓
Zhou et al. [96]	UAV			✓	✓	✓
Mahmood et al. [97]	UAV		✓		✓	✓

4.4. Conclusions

Our analysis of how AI can enhance the communication aspects of the swarm yielded interesting insights. First and foremost, we need to state that UAVs are the vehicle of choice for most researchers, with only one piece of bibliography focusing on USVs instead. This could again be due to the commercial success of aerial vehicles versus other types and their compatibility with arrays and other systems. Furthermore, most of the existing wireless communications already happen over the air, making UAVs another part of an existing ecosystem. In contrast, testing USVs and UUVs is more challenging due to their operational environment. While UAVs benefit from widespread terrestrial infrastructure—such as arrays and base stations—that support connectivity, similar infrastructure is largely absent on bodies of water like seas and lakes, making it difficult to ensure reliable communication between USVs. Things are even more complicated in the case of UUVs since water acts as the signal carrier, necessitating specialized equipment that is far less accessible than its terrestrial counterparts.

Regarding the communication aspects of the swarm, we have created Figure 6 to demonstrate results from Tables 6–8 comprehensively. We do not observe large deviations from bar to bar, showing that all aspects are researched for the most part. We also observe that the most researched topic is *Network Management and Optimization*, which makes sense if we consider that this category includes all the system-level strategies to dynamically allocate resources, maintain service quality, and optimize performance. *Channel and Propagation Modeling*, *Signal Quality and Enhancement*, and *Routing and Protocols* are the most equally researched; this could be due to the fact that the first two categories are both related to signals and how these are propagated, processed, and enhanced. Finally, we need to state that *Network Architecture and Topologies* is the least researched; therefore, it would be worth looking at the swarm from a network architecture perspective.

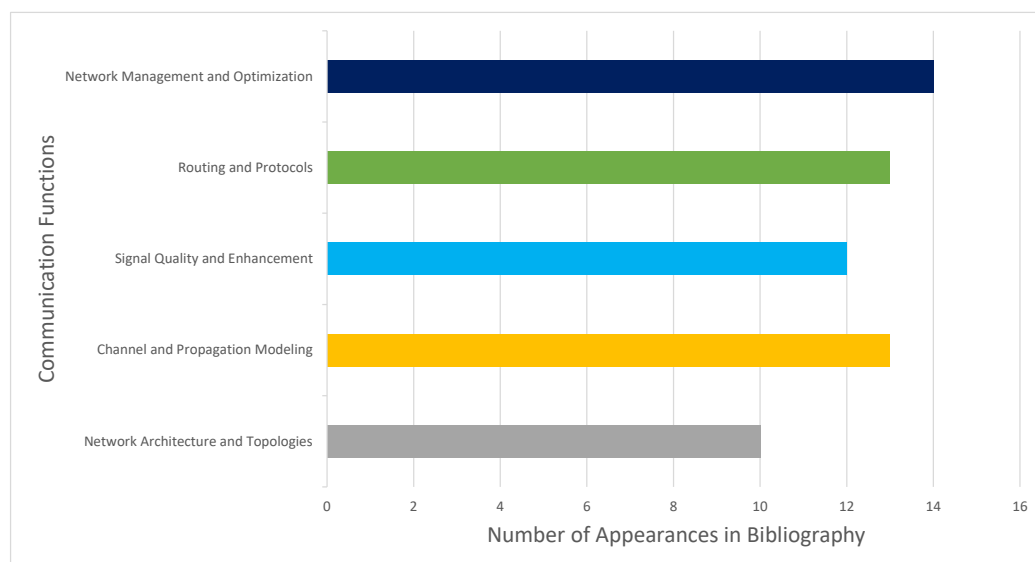


Figure 6. Appearances of each communication function in the surveyed literature.

5. AI-Driven Security Mechanisms

The final chapter of our survey revolves around the security posture and resilience of the swarm and how AI can help improve these areas. We follow the same methodology as before and try to identify operations related to security and resilience that have been enhanced by AI in the relevant literature. We have identified the following categories, which are also displayed in Figure 7:

- **Anomaly and Intrusion Detection:** Refer to the ability to identify abnormal behavior.
- **Data Integrity and Privacy:** These involve techniques to ensure that data are kept, transmitted, and processed in a secure manner.
- **Self Adaptability:** This refers to the swarm's ability to effectively respond to threats, faulty behavior, and generally dynamic internal and external changes.
- **Authentication Rules and Protocols:** These include rules and protocols that ensure that individual vehicles of the swarm are always authenticated.
- **Explainability:** AI, especially in security, should include methods to log decisions and provide an explanation.

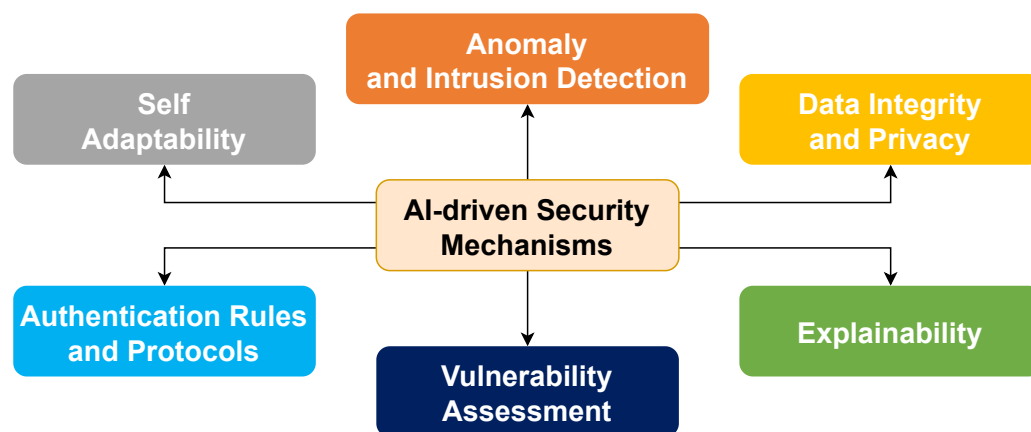


Figure 7. AI-driven security mechanisms.

We followed the same methodology as in the previous chapter, but this time, we only managed to create two chapters due to the lack of research:

- **Supervised, Federated, and Probabilistic Learning Models:** In this section, we deal with pieces of work which rely more on supervised NNs, FL, or probabilistic approaches.
- **Unsupervised, Hybrid, and Other Approaches:** Finally, this category includes the rest of the literature we found on the security aspects of the swarm.

5.1. Supervised, Federated, and Probabilistic Models

Supervised and FL methods form a significant portion of the existing research on UAV swarm security, particularly for tasks such as intrusion and anomaly detection, and swarm coordination. Table 9 displays this research alongside the security mechanisms tackled in each paper. Liu et al. [98] combine a multilayer perceptron with an XGBoost classifier to detect network intrusions within swarm communications. Similarly, da Silva et al. [99] incorporate both supervised and unsupervised techniques into an intrusion detection system, enhanced with FL to enable distributed and privacy-preserving threat detection. FL also plays a key role in the work of Challita et al. [100], where CNNs and RNNs are trained collaboratively across UAVs to support real-time authentication and identify cyber-physical threats in cellular-connected swarms. Along similar lines, Halli Sudhakara et al. [101] develop a federated intrusion detection system targeting GPS spoofing, further incorporating explainability through SHapley Additive exPlanations (SHAP) to make the model's behavior interpretable.

Another group of works focuses on anomaly detection using labeled data. Chouhan et al. [102] propose an NN-based framework for detecting anomalies, verifying object recognition, and ensuring data integrity. Their system integrates adaptive AI models and surveillance components for swarm defense scenarios. Expanding on this theme, Ahn et al. [103] introduce an anomaly detection method that uses principal component

analysis, clustering, and deep NNs to monitor kinematic flight data. This enables continuous tracking of drone behavior without relying on centralized control. An additional contribution comes from Nebe et al. [104], who address abnormal swarm dynamics by employing autoencoders, Gaussian mixture models, and CNNs to identify chaotic motion patterns. Their approach also includes a mechanism for rerouting drones based on trajectory corrections. Explainability within supervised models is also being explored. Gackowska-Katek et al. [105] propose a CatBoost-based model for predicting swarm disorganization resulting from intrusions. They try to relate collision avoidance and formation parameters to entropy, helping operators better interpret swarm instability.

Supervised and FL approaches offer promising tools for strengthening swarm security, particularly in intrusion detection and anomaly recognition. Supervised methods, as in [98,102], benefit from using labeled datasets and established classifiers, but their dependence on labeled data and centralized training often limits adaptability in environments where targeted adversarial actions occur. FL frameworks improve scalability and privacy by distributing learning between agents, yet they may be sensitive if attack patterns change and target different parts of the system. Finally, recent trends toward explainability, for example, refs. [101,105], rely on SHAP values for explainability, and we need to consider their efficiency in swarms employing diverse models to accomplish different tasks.

Table 9. Studies with approaches relying on supervised, federated, and probabilistic models.

Reference	Vehicle	Anomaly and Intrusion Detection	Self Adaptability	Data Integrity and Privacy	Authentication Rules and Protocols	Explainability	Vulnerability Assessment
Liu et al. [98]	Generic	✓					✓
Da Silva et al. [99]	UAV	✓		✓			
Challita et al. [100]	UAV	✓		✓	✓		
Halli Sudhakara and Haghne-gahdar [101]	UAV	✓		✓		✓	
Chouhan et al. [102]	UAV	✓	✓	✓			
Ahn et al. [103]	UAV	✓					
Nebe et al. [104]	UAV	✓	✓				
Gackowska-Katek and Cofta [105]	UAV	✓	✓			✓	

5.2. Unsupervised, Hybrid, and Other Methods

When labeled data are scarce or system conditions are highly dynamic, researchers turn to unsupervised or hybrid models to support security in UAV swarms. Table 10 displays the academic papers that rely on such methods. In [106], Wedaj Kibret proposes a property-based attestation framework that monitors system behavior to verify device integrity in large-scale swarms. This approach focuses on runtime properties rather than static binaries to detect impersonation and replay attacks. Identity verification and secure coordination are further addressed by Karmakar et al. in [107], who introduce SwarmAuth—a blockchain-

based authentication framework combining physically unclonable functions and K-means clustering. This method enables decentralized identity verification and cluster formation management. An interesting take on authentication can be found in [108], where Semenov et al. proposed a Fourier-based steganographic method for embedding UAV identifiers into ADS-B signals, enhancing resilience against spoofing and cyberattacks while preserving full compliance with automatic dependent surveillance–broadcast (ADS-B) standards. From a network architecture perspective, Grueber et al. [109] propose a software-defined networking (SDN) approach to swarm security. They integrate an RF classifier into the SDN controller to detect insider threats and mitigate traffic injection attacks via flow-level monitoring and secure routing protocols. Several studies take a broader or more strategic view. Tayyab et al. [110] examine the limitations of traditional drone security mechanisms, emphasizing risks posed by compromised swarm members. They propose a combination of AI-based security tools and regulatory safeguards to address ethical and legal considerations in swarm deployment. Finally, Phadke and Medrano [111] conduct a structured analysis of resiliency challenges in UAV swarms. They identify vulnerabilities across communication, mobility, task execution, and security layers and advocate for integrated, multi-module solutions that enhance resilience to jamming and intrusion.

These unsupervised and hybrid approaches broaden the security toolbox for UxV swarms by reducing dependence on labeled data and enabling more adaptive, decentralized defenses. Property-based attestation provides lightweight runtime verification, but may struggle with sophisticated adversaries capable of mimicking normal behavior. Blockchain-based frameworks, as in [107], improve decentralization and tamper-resistance, but raise concerns about latency, energy consumption, and scalability in resource-constrained swarms. SDN-based solutions centralize detection and control for better traffic management, but may introduce single points of failure in the system and require robust controller protection.

Table 10. Studies with approaches relying on unsupervised, hybrid, and other methods.

Reference	Vehicle	Anomaly and Intrusion Detection	Self Adaptability	Data Integrity and Privacy	Authentication Rules and Protocols	Explainability	Vulnerability Assessment
Wedaj Kibret [106]	Generic						✓
Karmakar et al. [107]	UAV			✓	✓		
Semenov et al. [108]	UAV				✓		
Guerber et al. [109]	UAV	✓	✓				
Tayyab et al. [110]	UAV	✓					✓
Phadke & Medrano [111]	UAV	✓	✓				

5.3. Conclusions

Finishing our analysis on the security mechanisms of the swarm and how they could be improved with AI, we gather our results in Figure 8. Here, we observe that anomaly and intrusion detection against attacks is a popular research choice. A representative type of attack that frequently affects UxVs is jamming. Jamming interferes with the communication frequencies of UxVs, rendering them unable to receive commands from their operator and

therefore complete their objective. This technique has been used extensively against aerial vehicles in real-world applications; therefore, applying countermeasures against it is a very compelling area of research. For example, in [112], Naous et al. propose the combination of RL and non-terrestrial networks to mitigate any jamming attempts against UAVs, as well as ways to ensure that data integrity and privacy are preserved. Although we see little effort to ensure the explainability of such methods, which is important when AI makes decisions on anomaly classification or acts on its own to adapt to these anomalies and new circumstances, if we look at [101,105], they both rely on SHAP analysis to identify key contributing features. SHAP values can prove helpful in explaining ML decisions, but there is also a certain level of untrustworthiness, as demonstrated in [113], and as ML models become more complex with more diverse inputs, the need for more dynamic and robust methods increases.

Furthermore, as swarms tend to grow in size and complexity, efficient and dynamic authentication schemes will be more needed to ensure that no external vehicles have infiltrated the formation. While ADS-B exists, it is a protocol that lacks encryption and strict authentication, creating the need for supplementary and robust mechanisms such as those proposed in [114]. Finally, since UxVs integrate commercial-off-the-shelf (COTS) software and hardware components, which might not be tested for zero-day exploits or other exploitable behaviors, ways to rapidly and efficiently test them will be vital in the future.

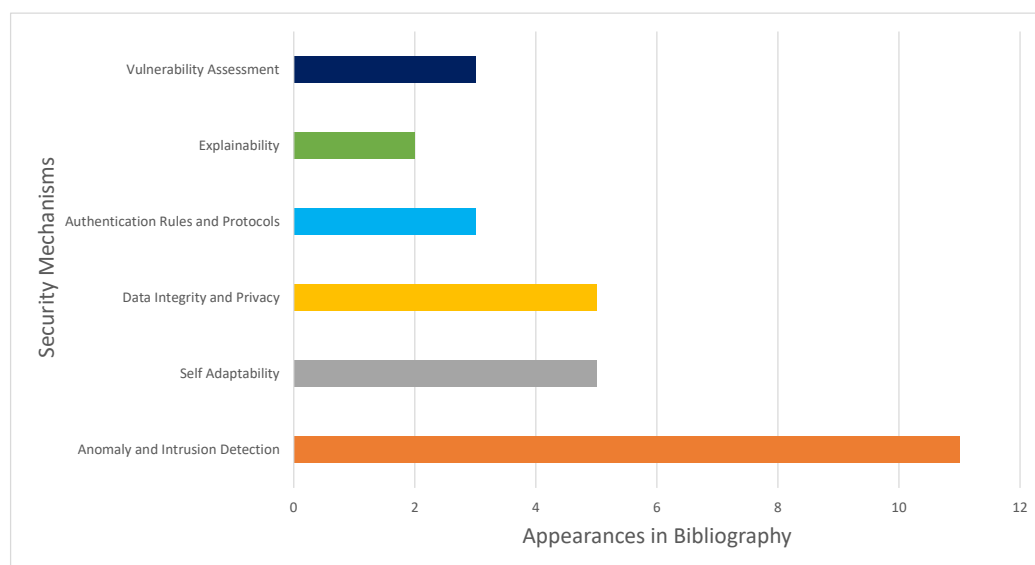


Figure 8. Appearances of each security mechanism in the surveyed literature.

6. Discussion

Our survey on UxV swarms has mapped the literature landscape and highlighted strengths and weaknesses across three dimensions:

- Type of vehicle;
- Swarm aspects;
- Type of AI.

Regarding vehicles, Figure 9 illustrates the distribution of vehicle types in the bibliography of this review. By doing this, we wanted to highlight the general research trend. As we observe, there is a preference for UAVs, while ground and underwater swarms remain underexplored. We see research opportunities in adapting existing AI-based approaches to these environments, as shown in works on surface vessels such as [39,46] and underwater

systems [115,116]. Ground swarms are particularly scarce, and their integration with AI remains an open avenue.

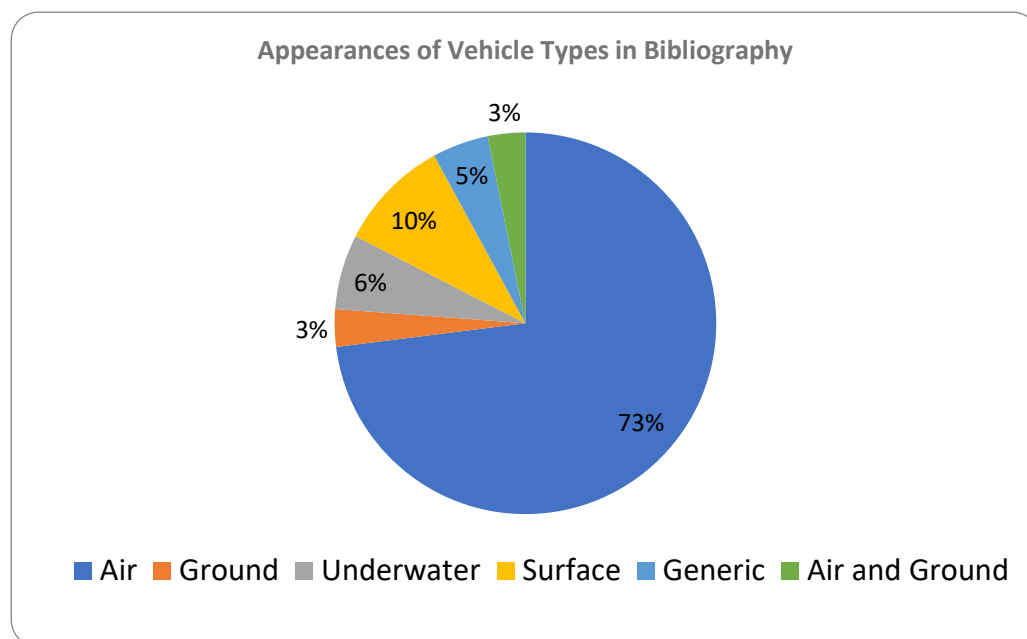


Figure 9. Appearances of different types of vehicles in the surveyed literature.

Across swarm aspects, our review in Sections 3–5 shows that intelligence and operations such as path planning, formation control, and obstacle avoidance are well studied. Less attention is given to task and resource optimization, although these underpin the swarm’s capabilities and behaviors. In communication, the focus on UAVs and reliance on terrestrial networks offer limited room for future research. Works such as [112,117] prompt us to move one step further and integrate aerial vehicles in non-terrestrial networks (NTNs). NTNs could also prove extremely beneficial for swarms of UGVs or water vehicles, which operate in remote locations with limited coverage by terrestrial networks. Furthermore, developing water-based communication for UUVs could open new research directions as described in [118]. Security research is largely based on intrusion detection, with some steps toward self-adaptability. However, proactive measures such as vulnerability assessment, authentication, and explainability become critical as AI/ML integration deepens. These insights are summarized in Figure 10.

In terms of AI, heuristic algorithms such as ACO, PSO, and ABC remain widely used, often in hybrid approaches with modern ML. RL and FL stand out as particularly suitable for swarms due to their agent-based nature. We need to state that applying RL algorithms in the real world is challenging though, as described in [119]. Valuable insight in this regard is also provided by Salvato et al. in [120] and Tobin et al. in [121], where they show that RL policies trained in simulation often fail in the real world due to the “real world gap”, i.e., unmodeled dynamics, noise, and limited variability in simulators, and that techniques such as domain randomization are needed to partially mitigate these transfer failures. Looking ahead, generative AI and LLMs represent a promising frontier. Recent studies [122–128] demonstrate their potential for anomaly detection, incident response, and reasoning tasks. We envision LLMs supporting mission planning, self-adaptability, and resilience against disruptions. We could also consider custom and lightweight LLMs embedded in individual vehicles, complemented by larger models overseeing the swarm.



Figure 10. Research directions on swarm technologies.

From our synthesis, several research gaps emerge. These include the following:

- Expansion beyond UAVs to ground and underwater swarms or combinations of different vehicles in the same swarm.
- Facing task and resource optimization as standalone swarm functions.
- Development of communication techniques and coverage expansion to environments with limited coverage by terrestrial means.
- Adoption of proactive and AI-driven security strategies with a focus on explainability.
- Integration of LLMs into swarm decision-making.

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Abbreviations

The following abbreviations are used in this manuscript:

ACO	Ant Colony Optimization
ADS-B	Automatic Dependent Surveillance–Broadcast
AI	Artificial Intelligence
CNNs	Convolutional Neural Networks
COTS	Commercial-Off-The-Shelf
DDPG	Deep Deterministic Policy Gradient
DDQL	Double Deep Q-Learning
DL	Deep Learning
DOAJ	Directory of Open Access Journals
DQN	Deep Q-Networks
DRL	Deep Reinforcement Learning
FANETs	Flying Ad Hoc Networks
FL	Federated Learning
GAI	Generative Artificial Intelligence
GAN	Generative Adversarial Network
IoD	Internet of Drones
IoT	Internet of Things
KNN	K-Nearest Neighbors
LLMs	Large Language Models
MARL	Multi-Agent RL
MDPI	Multidisciplinary Digital Publishing Institute
ML	Machine Learning
MIMO	Multiple-Input and Multiple-Output
NNs	Neural Networks
NOMA	Non-Orthogonal Multiple Access
NTN	Non-Terrestrial Network
PDR	Packet Delivery Ratio
PPO	Proximal Policy Optimization
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PSO	Particle Swarm Optimization
QoS	Quality of Service
RL	Reinforcement Learning
RNNs	Recurrent Neural Networks
SDN	Software-Defined Networking
SHAP	SHapley Additive exPlanations
SI	Swarm Intelligence
SINR	Signal-to-Interference-Plus-Noise Ratio
SotA	State of the Art
UAVs	Uncrewed Aerial Vehicles
UGVs	Uncrewed Ground Vehicles
USVs	Uncrewed Surface Vehicles
UUVs	Uncrewed Underwater Vehicles
UxV	Uncrewed Vehicle
VLC	Visible Light Communication

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